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(54) **SYSTEMS AND METHODS FOR
NON-EPITAXIAL HIGH
SCHOTTKY-BARRIER HETEROJUNCTION
SOLAR CELLS**

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ABSTRACT

Systems and methods of non-epitaxial high Schottky barriers heterojunction solar cells are described. The high Schottky barriers heterojunction solar cells are formed using non-epitaxial methods to reduce fabrication costs and improve scalability.

8 Claims, 8 Drawing Sheets

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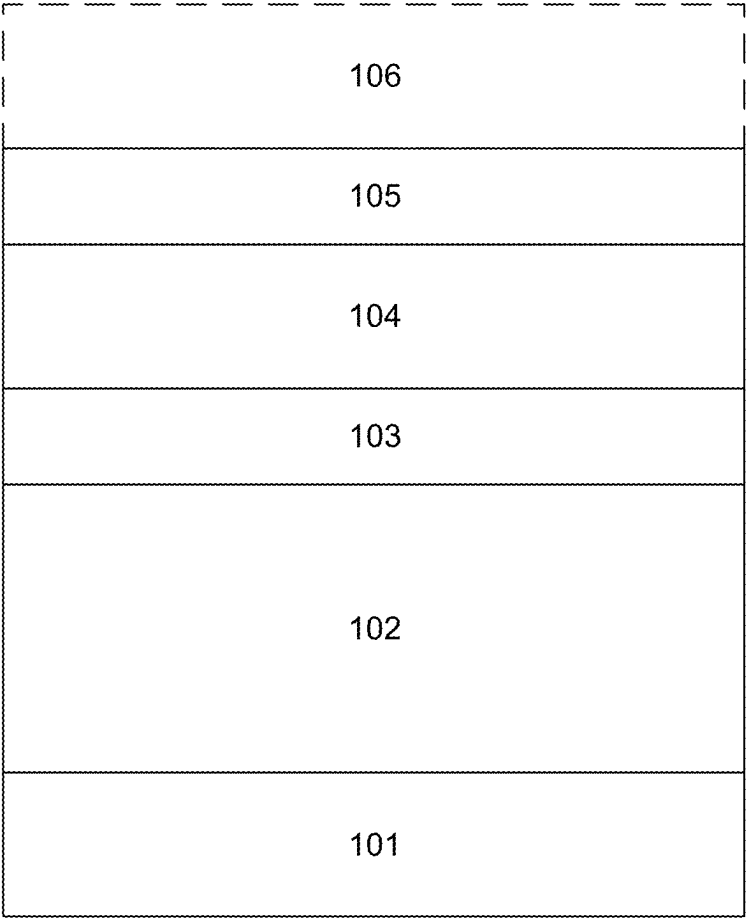


FIG. 1

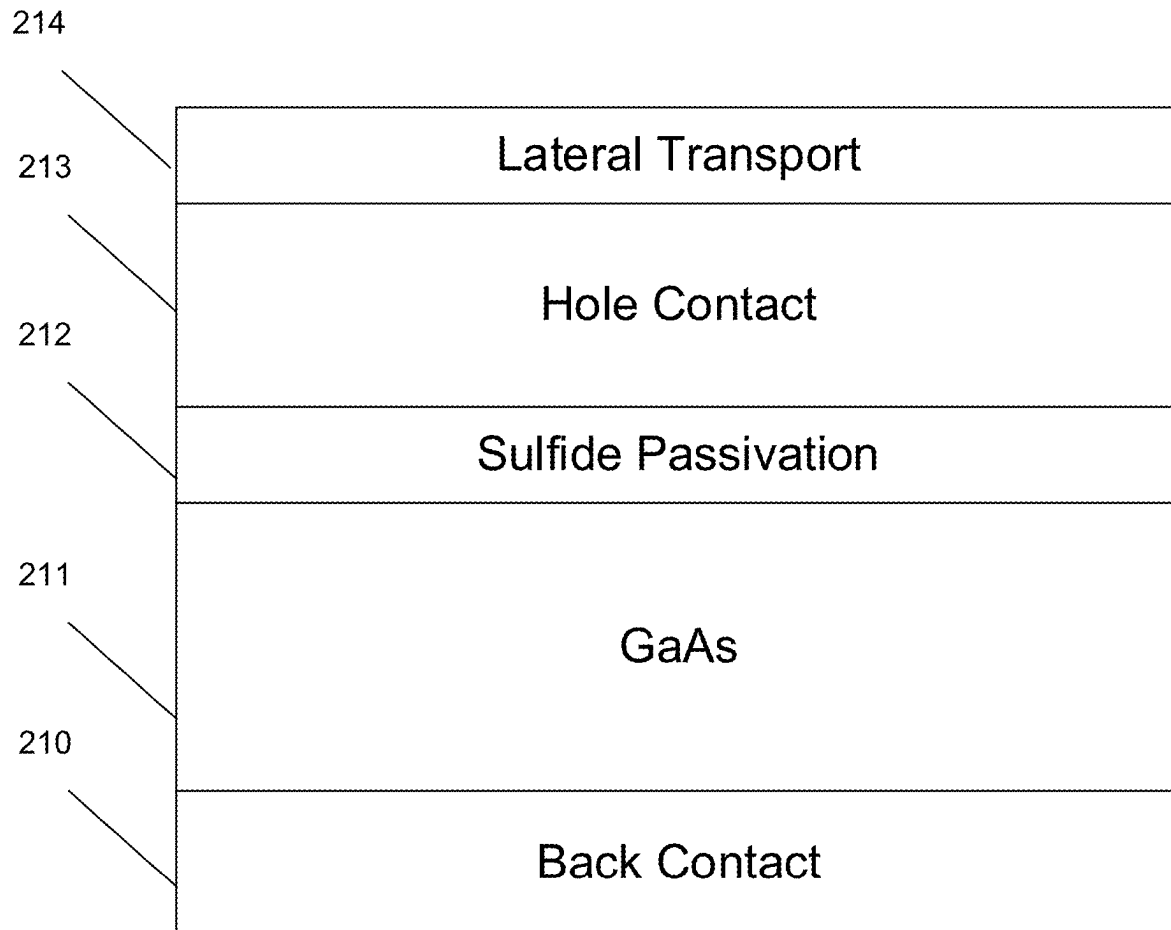
*FIG. 2*

FIG. 3A

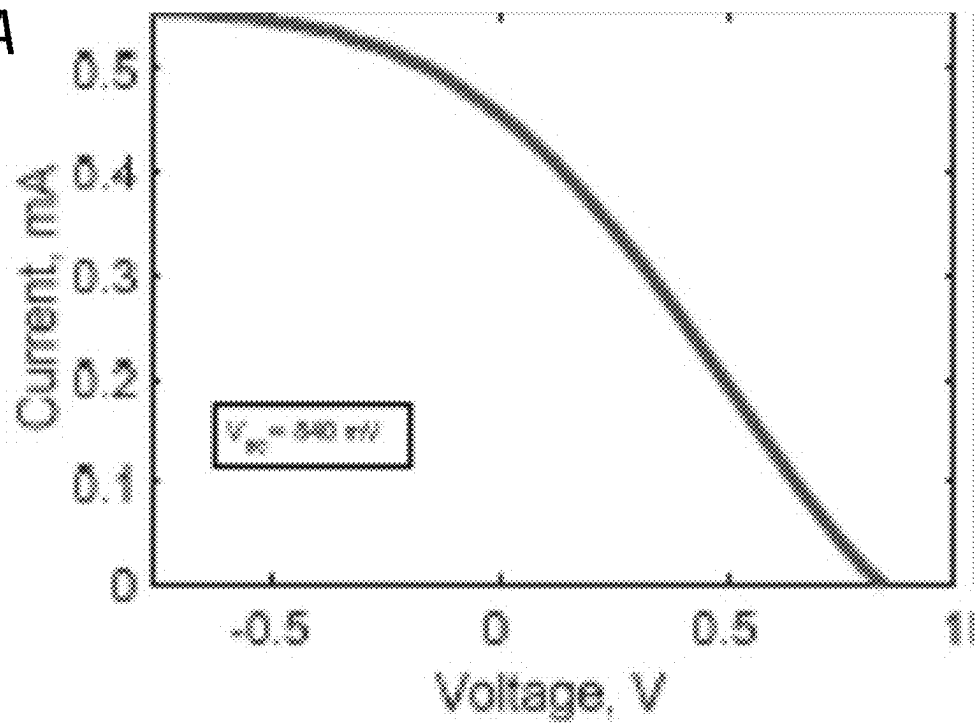
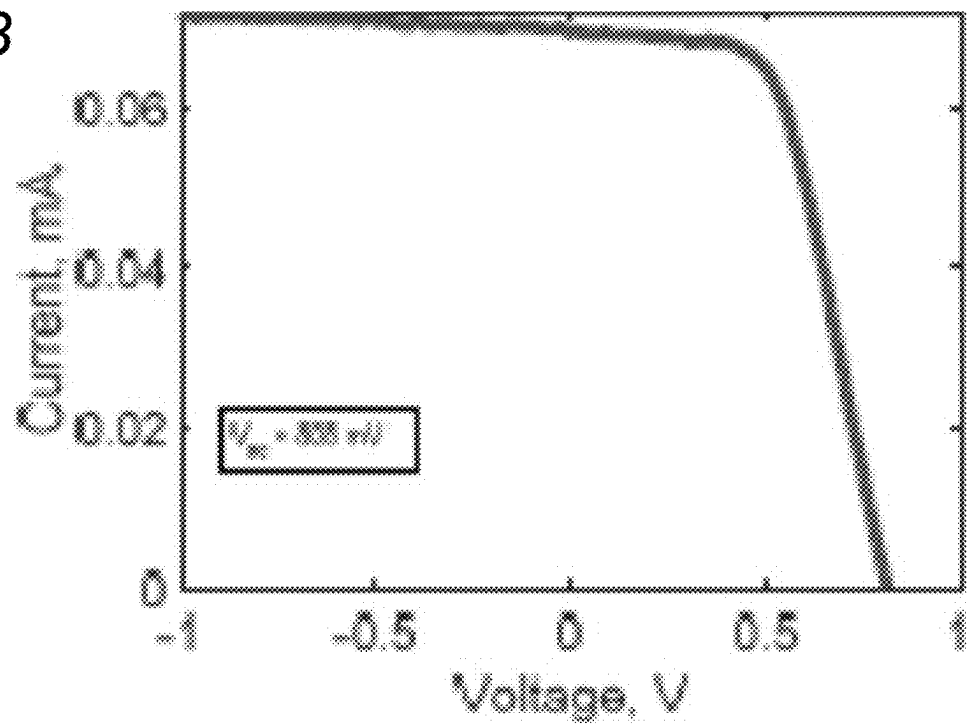


FIG. 3B



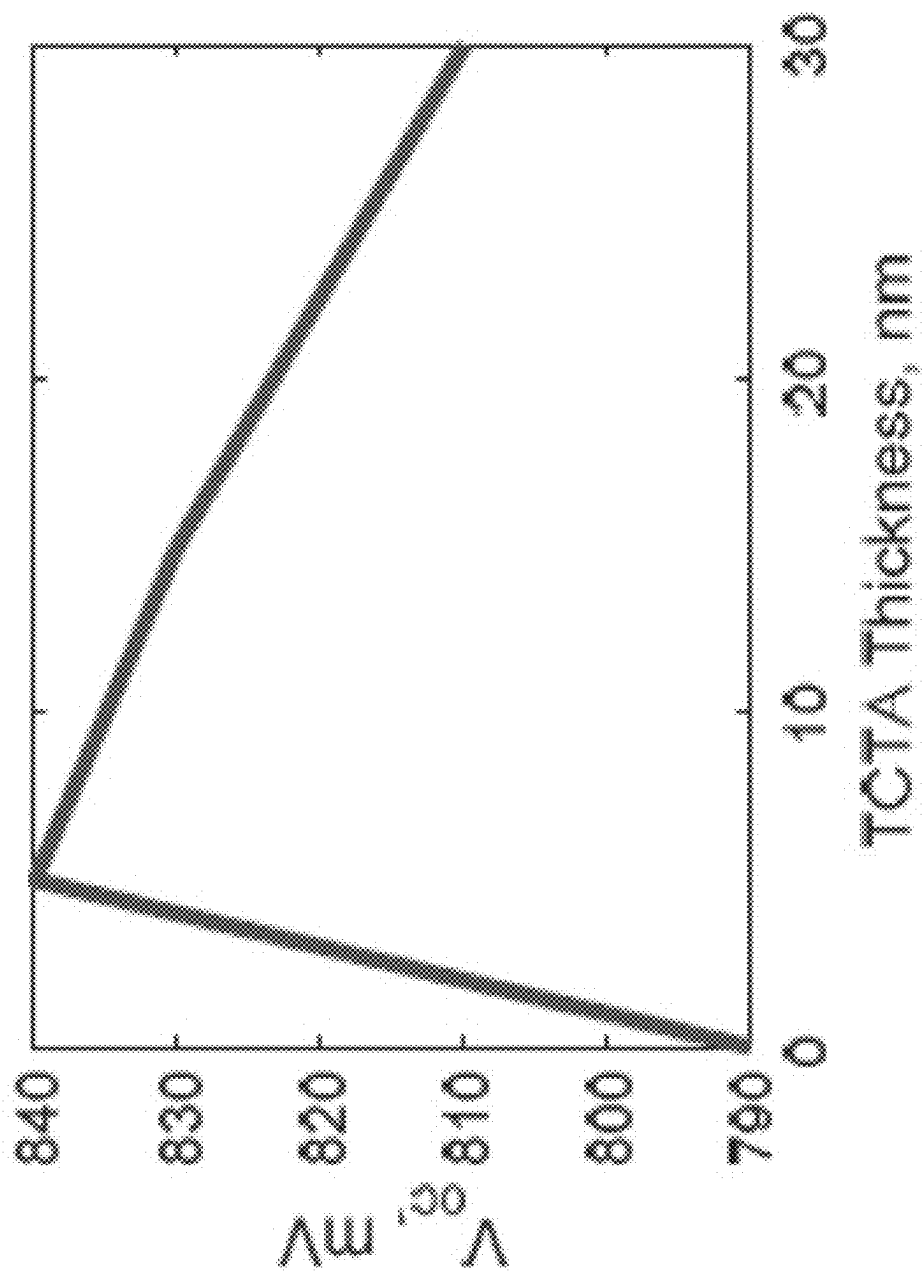


FIG. 4

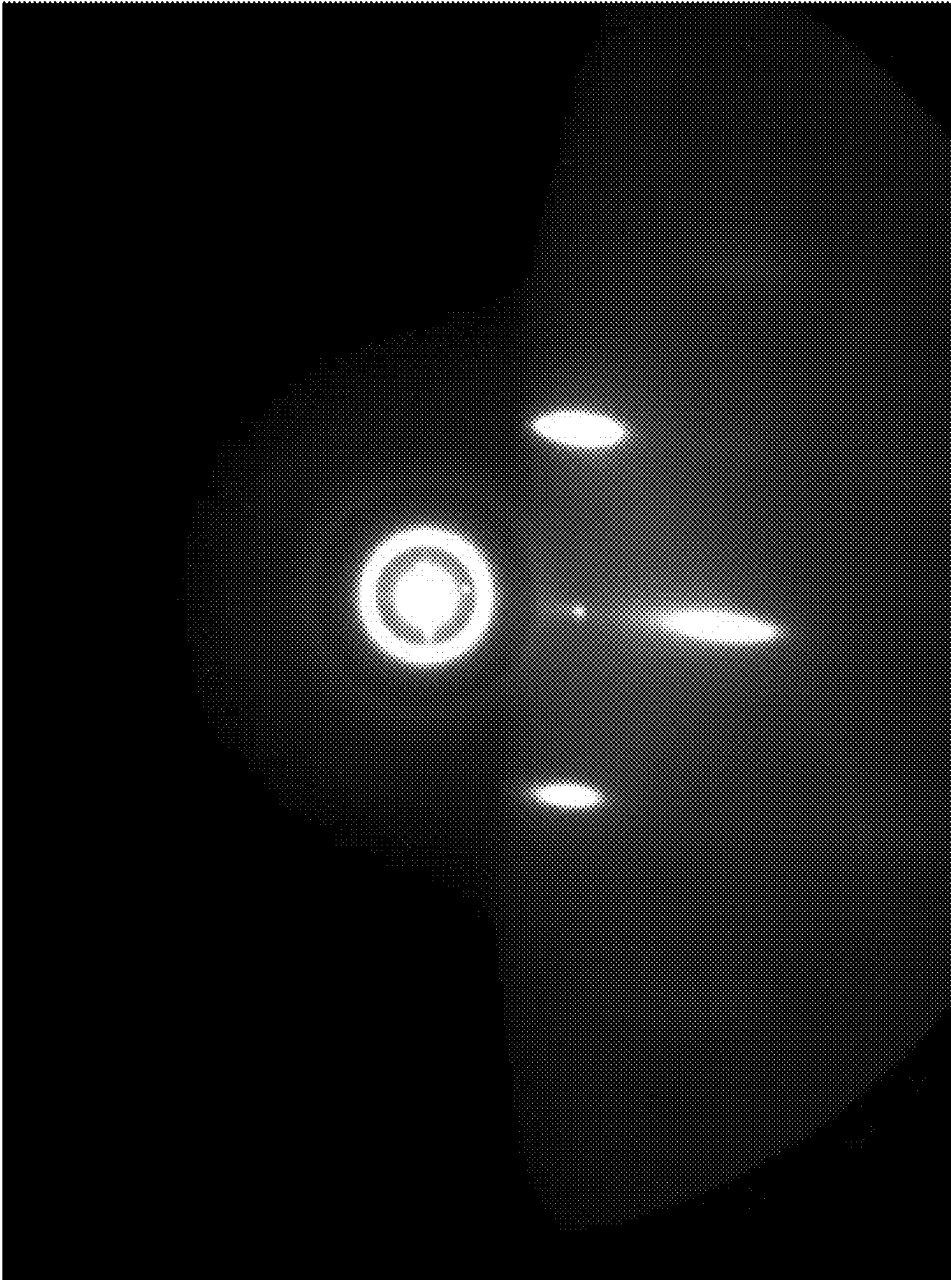


FIG. 5

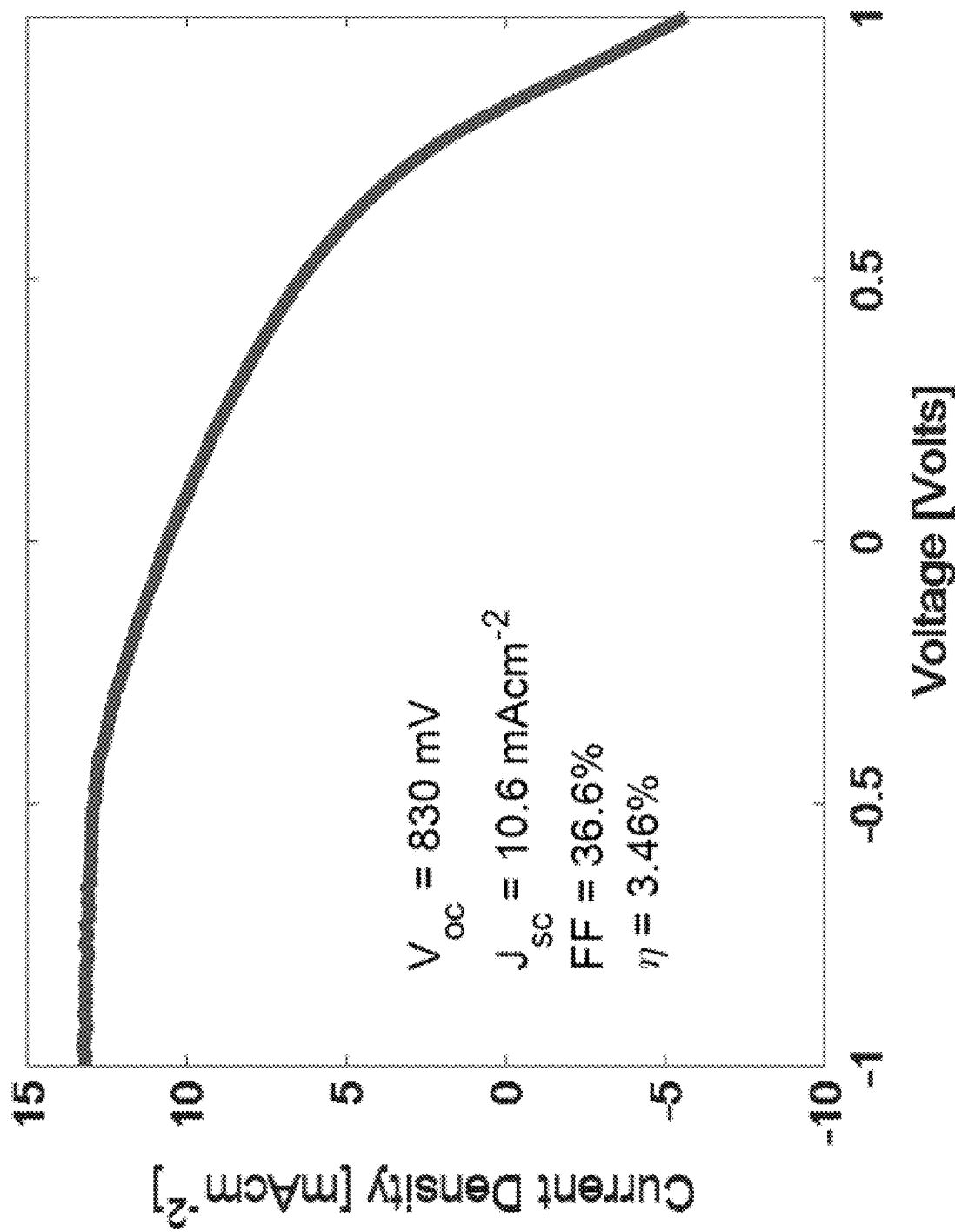


FIG. 6

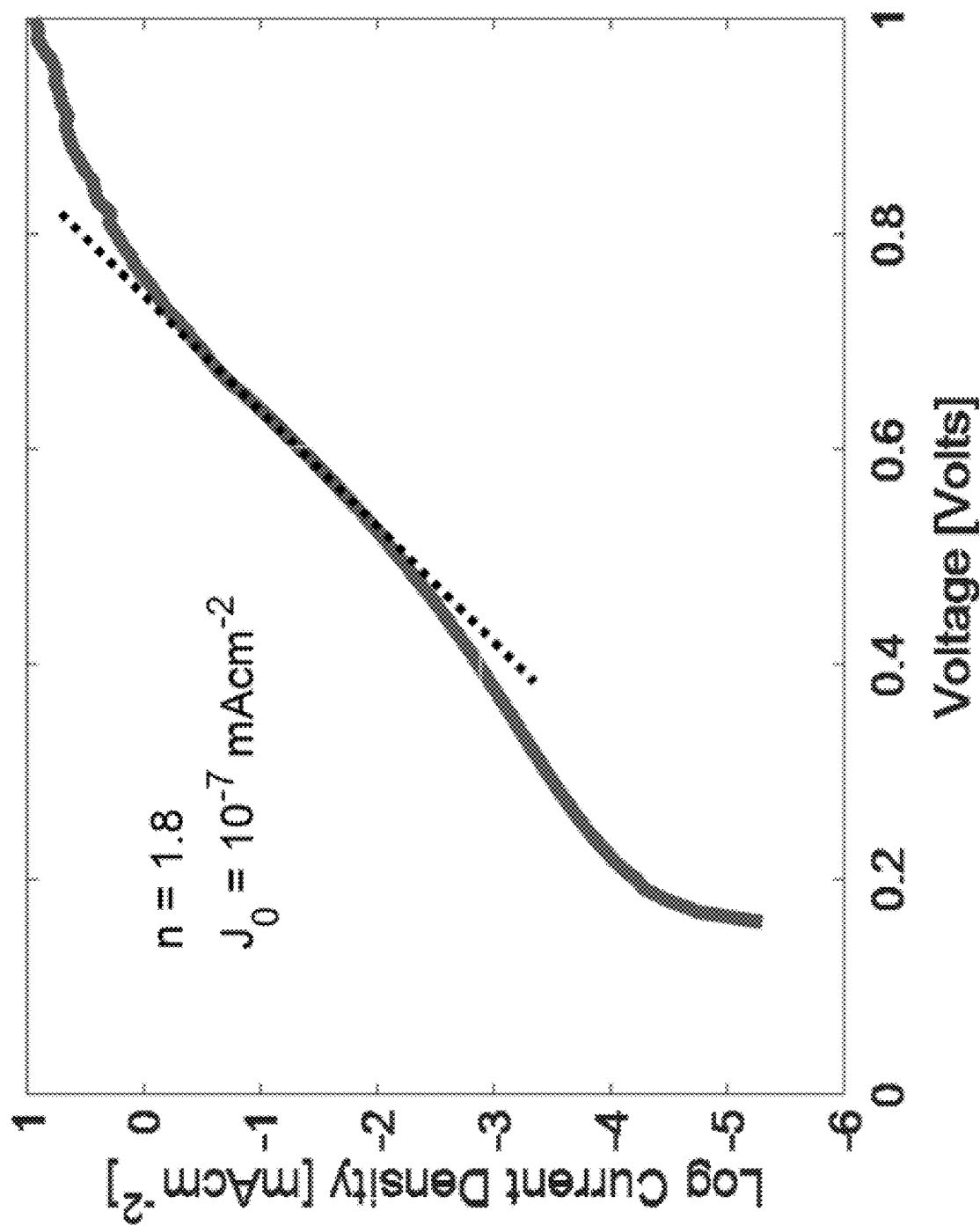
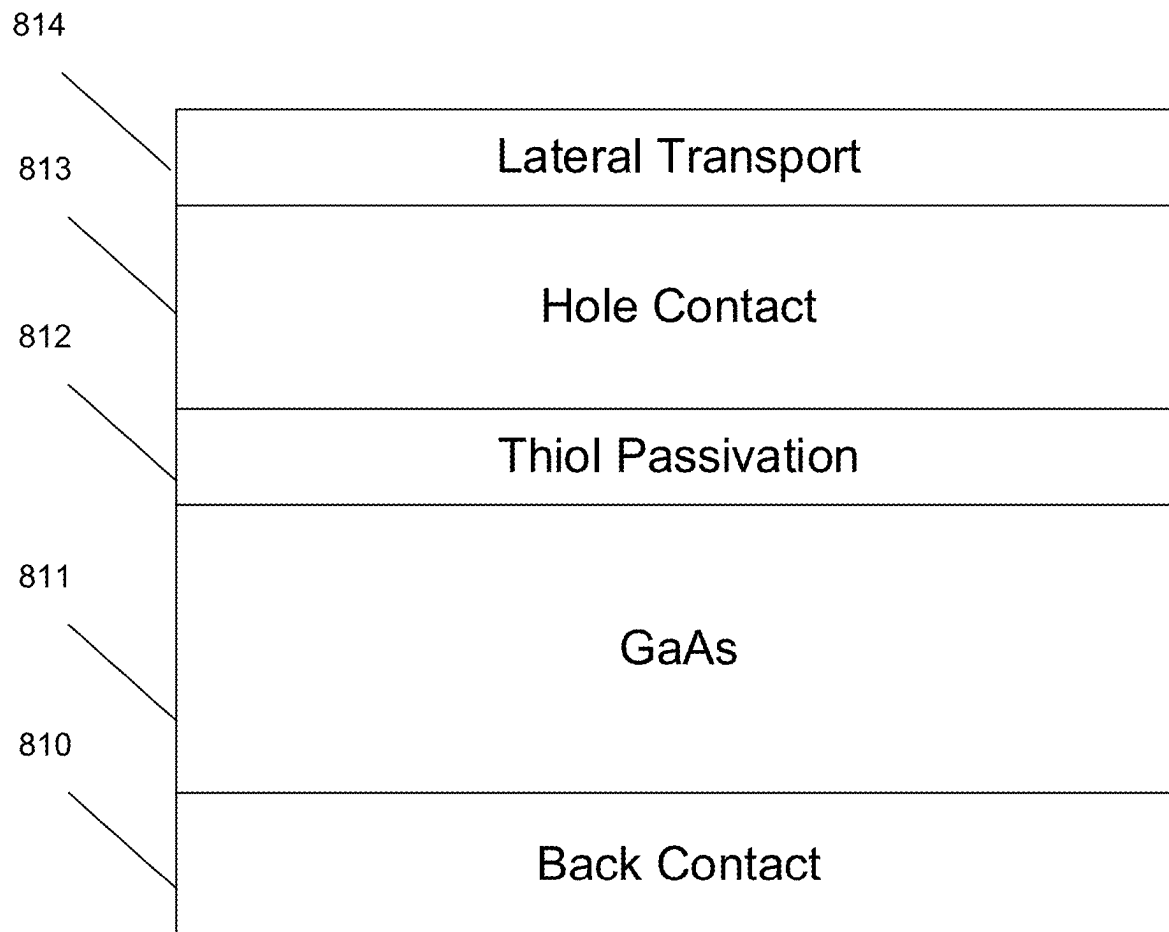


FIG. 7

*FIG. 8*

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SYSTEMS AND METHODS FOR NON-EPITAXIAL HIGH SCHOTTKY-BARRIER HETEROJUNCTION SOLAR CELLS

CROSS-REFERENCE TO RELATED APPLICATIONS

The current application claims the benefit of and priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/038,546 entitled "High Schottky-Barrier GaAs Solar Cells" filed Jun. 12, 2020. The disclosure of U.S. Provisional Patent Application No. 63/038,546 is hereby incorporated by reference in its entirety for all purposes.

STATEMENT OF FEDERALLY SPONSORED RESEARCH

This invention was made with government support under Grant No. EEC1041895 awarded by the National Science Foundation. The government has certain rights in the invention.

FIELD OF THE INVENTION

The present invention generally relates to systems and methods for non-epitaxial heterojunction solar cells with high Schottky barriers; and more particularly to systems and methods that passivate heterojunction solar cell surfaces and make electrical contacts to them to achieve high Schottky barriers.

BACKGROUND

High-efficiency multijunction devices may use multiple bandgaps, or junctions, that are tuned to absorb a specific region of the solar spectrum to create solar cells having high efficiencies. The maximum theoretical efficiency that a single-bandgap solar cell can achieve with non-concentrated sunlight is lower, primarily because of the broad distribution of solar emitted photons. This limiting efficiency, known as the Shockley-Queisser limit, arises from the fact that the open-circuit voltage (V_{oc}) of a solar cell is limited by the bandgap of the absorbing material and that photons with energies below the bandgap are not absorbed. Photons that have energies greater than the bandgap are absorbed, but the energy greater than the bandgap is lost as heat.

Multijunction devices can use a high-bandgap top cell to absorb high-energy photons while allowing the lower-energy photons to pass through. A material with a slightly lower bandgap is then placed below the high-bandgap junction to absorb photons with slightly less energy (longer wavelengths). Multijunction cells can use two or more absorbing junctions, and the theoretical maximum efficiency increases with the number of junctions. Early research into multijunction devices leveraged the properties of semiconductors comprised from elements in the III and V columns of the periodic table.

Traditional multijunction III-V cells are assembled in an epitaxial monolithic stack with subcells connected in series through by tunnel junctions. Constructing a multijunction cell in a monolithic stack results in material constraints, and fabricating such devices is facilitated if the individual layers of the subcells have compatible atomic lattice positions and are lattice matched.

BRIEF SUMMARY OF THE INVENTION

Systems and methods in accordance with various embodiments of the invention enable the design and/or implemen-

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tation of non-epitaxial high Schottky barriers heterojunction solar cells. In many embodiments, heterojunction solar cells include semiconductor materials from the III and V columns of the periodic table including (but not limited to): gallium arsenide (GaAs). Several embodiments include n-type doped GaAs solar cells (n-GaAs) and p-type doped GaAs solar cells (p-GaAs). Heterojunction solar cells in accordance with many embodiments are formed using non-epitaxial methods. Several embodiments provide that non-epitaxial heterojunction solar cells reduce fabrication costs and improve scalability.

Some embodiments passivate heterojunction solar cell surfaces and also make electrical contacts to them to achieve high Schottky barriers. Several embodiments provide that the passivation layers can include organic layers. In some embodiments, layers with bulky tails can be deposited on top of the passivation layers to protect the passivation layers and the solar cell surfaces. Certain embodiments provide that the bulky tails are sufficiently short so that electrons and/or holes can pass through. In a number of embodiments, transparent conductor layers can be deposited on top of the bulky tail layers. In many embodiments, open-circuit voltage (V_{oc}) of high Schottky barrier heterojunction solar cells can range from about 20 mV to about 900 mV. The efficiency of the heterojunction solar cells in accordance with several embodiments is at least 20%.

One embodiment of the invention includes a non-epitaxial heterojunction solar cell, comprising at least one back contact; at least one semiconductor substrate, where the semiconductor is an n-type doped GaAs or a p-type doped GaAs; at least one passivation layer, wherein the passivation layer passivates one surface of the semiconductor substrate; at least one contact layer, wherein the contact layer is a hole transport layer or an electron transport layer, and the contact layer is on top of the passivation layer; and at least one electrode modified layer, wherein the electrode modified layer is on top of the contact layer.

In another embodiment, the back contact is an ohmic contact comprising Cu and Ge.

In a further embodiment, the passivation layer is an organic film comprising at least one thiol group.

In still another embodiment, the organic film comprises octanethiol (OT), 3,3,4,4,5,5,6,6,7,7,8,8,8-tridecafluoro-1-octanethiol (FOT), pentafluorothiophenol (PFBT), dithiothreitol (DTT), or cysteine (Cys).

In a yet further embodiment, the passivation layer is an organic film comprising N-heterocyclic-carbene.

In yet another embodiment, the organic film comprises 1,3-Diisopropylimidazolium Hydrogencarbonate (NHC).

In a still further embodiment, the passivation layer comprises at least one sulfide ion.

In yet another embodiment, the passivation layer comprises ammonium sulfide.

In a further embodiment again, the hole transport layer comprises one of poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS), 2,2',7,7'-Tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (SPIRO), SPIRO doped with 5 wt % Tris(pentafluorophenyl)borane, tris(4-carbazoyl-9-ylphenyl)amine (TCTA), TCTA doped with 5 wt % Tris(pentafluorophenyl)borane, polystyrene (PS), 4,4'-Bis(N-carbazoyl)-1,1'-biphenyl (CBP), copper (I) iodide (CuI) dissolved in acetonitrile, polyvinyl alcohol (PVA), copper thiocyanate (CuSCN), 1,1-Bis[(di-4-tolylamino)phenyl]cyclohexane (TAPC), TAPC doped with 5 wt % Tris(pentafluorophenyl)borane, poly(triaryl amine) (PTAA), PTAA doped with 5 wt % Tris(pentafluorophenyl)borane, nickel oxide (NiO) nanoparticles, or NiO film.

In a further embodiment again, the electron transport layer comprises one of tin oxide (SnO_2) nanoparticles, aluminum doped zinc oxide (AZO) nanoparticles, or C60.

In a still yet further embodiment, the semiconductor substrate is an n-type doped GaAs and the electrode modified layer comprises tungsten trioxide (WO_3) or molybdenum trioxide (MoO_3).

In another additional embodiment, the semiconductor substrate is an p-type doped GaAs and the electrode modified layer comprises lithium fluoride (LiF) or caesium oxide (Cs_2O).

A still further embodiment includes a lateral current transport layer, where the lateral current transport layer comprises poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS), indium tin oxide (ITO), or silver nanowires.

In a still yet further embodiment, the open circuit voltage of the solar cell is at least 830 mV.

Still another additional embodiment includes a method of fabricating a non-epitaxial heterojunction solar cell comprising: providing a semiconductor substrate, wherein the semiconductor substrate does not have any epitaxial layers; depositing a back contact layer on a first side of the semiconductor substrate; passivating a second side of the semiconductor substrate with at least one passivation layer; coating the passivation layer with at least one contact layer, wherein the contact layer is a hole transport layer or an electron transport layer; and depositing at least one electrode modified layer on the contact layer.

In a further embodiment again, the semiconductor substrate is an n-type doped GaAs or a p-type doped GaAs.

In yet another embodiment, the back contact is an ohmic contact comprising Cu and Ge and the back contact is deposited by thermal evaporation.

In a still further additional embodiment, the passivation layer is an organic film comprising at least one thiol group and the organic film comprises octanethiol (OT), 3,3,4,4,5,5,6,6,7,7,8,8,8-tridecafluoro-1-octanethiol (FOT), pentafluorothiophenol (PFBT), dithiothreitol (DTT), or cysteine (Cys).

In another yet further embodiment, the passivation layer is an organic film comprising N-heterocyclic-carbene and the organic film comprises 1,3-Diisopropylimidazolium Hydrogencarbonate (NHC).

In yet another embodiment again, the passivation layer comprises ammonium sulfide.

In still yet another further embodiment, the hole transport layer comprises one of poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS), 2,2',7,7'-Tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (SPIRO), SPIRO doped with 5 wt % Tris(pentafluorophenyl)borane, tris(4-carbazoyl-9-ylphenyl)amine (TCTA), TCTA doped with 5 wt % Tris(pentafluorophenyl)borane, polystyrene (PS), 4,4'-Bis(N-carbazoyl)-1,1'-biphenyl (CBP), copper (I) iodide (CuI) dissolved in acetonitrile, polyvinyl alcohol (PVA), copper thiocyanate (CuSCN), 1,1-Bis[(di-4-tolylamino)phenyl]cyclohexane (TAPC), TAPC doped with 5 wt % Tris(pentafluorophenyl)borane, poly(triaryl amine) (PTAA), PTAA doped with 5 wt % Tris(pentafluorophenyl)borane, nickel oxide (NiO) nanoparticles, or NiO film.

In still another further embodiment again, the electron transport layer comprises one of tin oxide (SnO_2) nanoparticles, aluminum doped zinc oxide (AZO) nanoparticles, or C60.

In another further additional embodiment, the semiconductor substrate is an n-type doped GaAs and the electrode modified layer comprises tungsten trioxide (WO_3) or molybdenum trioxide (MoO_3).

In yet a further embodiment again, the semiconductor substrate is an p-type doped GaAs and the electrode modified layer comprises lithium fluoride (LiF) or caesium oxide (Cs_2O).

A still further embodiment again includes depositing a lateral current transport layer on the electrode modified layer, wherein the lateral current transport layer comprises poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS), indium tin oxide (ITO), or silver nanowires.

Additional embodiments and features are set forth in part in the description that follows, and in part will become apparent to those skilled in the art upon examination of the specification or may be learned by the practice of the disclosure. A further understanding of the nature and advantages of the present disclosure may be realized by reference to the remaining portions of the specification and the drawings, which forms a part of this disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The description will be more fully understood with reference to the following figures, which are presented as exemplary embodiments of the invention and should not be construed as a complete recitation of the scope of the invention, wherein:

FIG. 1 illustrates a non-epitaxial heterojunction solar cell structure in accordance with an embodiment of the invention.

FIG. 2 illustrates a GaAs/TCTA/ WO_3 heterojunction solar cell structure in accordance with an embodiment of the invention.

FIGS. 3A and 3B illustrate IV curves of GaAs/TCTA/ WO_3 heterojunction solar cell with different TCTA thickness in accordance with an embodiment of the invention.

FIG. 4 illustrates V_{oc} of GaAs/TCTA/ WO_3 heterojunction solar cells with various TCTA thickness in accordance with an embodiment of the invention.

FIG. 5 illustrates RHEED image of passivated GaAs after annealing in air at 300° C. in accordance with an embodiment of the invention.

FIG. 6 illustrates a light IV curve for GaAs/TCTA/ WO_3 heterojunction solar cell that achieves 830 mV V_{oc} in accordance with an embodiment of the invention.

FIG. 7 illustrates a dark IV curve for GaAs/TCTA/ WO_3 heterojunction solar cell in accordance with an embodiment of the invention.

FIG. 8 illustrates a GaAs/Thiol/ WO_3 heterojunction solar cell structure in accordance with an embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

Turning now to the drawings, non-epitaxial heterojunction solar cells with high Schottky barriers in accordance with various embodiments are illustrated. Many embodiments implement heterojunction solar cells with semiconductor materials from the III and V groups of the periodic table. Semiconductor materials of heterojunction solar cells include (but are not limited to): gallium arsenide (GaAs). Several embodiments include n-type doped GaAs solar cells (n-GaAs) and p-type doped GaAs solar cells (p-GaAs). Heterojunction solar cells in accordance with many embodi-

ments are formed without any epitaxial processing. Some embodiments provide that heterojunction solar cells are formed without external doping on a bulk wafer. Several embodiments provide that non-epitaxial heterojunction solar cells reduce fabrication costs and improve scalability.

Many embodiments provide non-epitaxial processes to form high Schottky barrier solar cells. Some embodiments passivate heterojunction solar cell surfaces and also make electrical contacts to them to achieve high Schottky barriers. Several embodiments provide that the passivation layers can include organic layers. In certain embodiments, the organic layers can bond to the solar cell surfaces and passivate them. In some embodiments, the passivation layers can be organic layers containing at least one thiol group. The thiol containing passivation layers can make high-barrier Schottky junctions in accordance with many embodiments. Several embodiments provide that the passivation layers can be organic layers containing N-heterocyclic-carbene. The passivation layers in accordance with certain embodiments can be organic layers containing sulfide ions including (but not limited to) ammonium sulfide.

In some embodiments, electron and/or hole transport contacts can be deposited on top of the passivation layers to protect the passivation layers and the solar cell surfaces. In several embodiments, the electron/hole contact layers contain bulky tails with a certain length so that electrons and/or holes can pass through. Certain embodiments provide that electron and/or hole transport contact layers should not destroy the semiconductor and the passivation layer interface. In a number of embodiments, electron and/or hole transport contacts can be solution deposited nanoparticles, solution deposited organics, evaporated metal oxides, and organics. Some embodiments provide that the organics may allow deposition of metals to make high-barrier Schottky contacts.

In a number of embodiments, electrode modified layers can be deposited on top of electron/hole transport layers. The electrode modified layers in accordance with several embodiments can set the work function of the contacts. Examples of the electrode modified layers for n-GaAs include (but are not limited to): tungsten trioxide (WO_3), molybdenum trioxide (MoO_3). Examples of electrode modified layers for p-GaAs include (but are not limited to): lithium fluoride (LiF), caesium oxide (Cs_2O).

In several embodiments, lateral current transport layers can be deposited on top of the contacts. The lateral current transport layers in accordance with some embodiments can be transparent conductors. Examples of lateral current transport layers include (but are not limited to): poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS), indium tin oxide (ITO), silver nanowires.

Many embodiments provide that open-circuit voltage (V_{oc}) of high Schottky barrier heterojunction solar cells can range from about 20 mV to about 900 mV. In some embodiments, V_{oc} can be about 830 mV. The efficiency of the heterojunction solar cells in accordance with several embodiments can be about 20%.

Heterojunction Solar Cells

The leveled cost of energy of novel photovoltaic technology may need to be competitive with silicon and other power generation technologies in order to compete with silicon and contribute to fighting global warming. In order to grow quickly from pilot-plant scale to mass-manufacturing it may be necessary that the novel technology have a low capital intensity so that profit from selling devices can be quickly converted into new factories to manufacture devices. The capital intensity of silicon photovoltaics can be

dominated by converting sand into semiconductor-grade wafers which may suggest two strategies for achieving high growth-rates. First, avoid using wafers. This is the strategy explored by perovskites, CdTe, CIGS, and organic photovoltaics which implement non-epitaxial heterojunction devices. (See, e.g., Q. Jiang, et al., *Nat. Photonics*, 2019, 13, 7, 460-466; L. Meng, et al., *Science*, 2018, 361, 6407, 1094-1098; the disclosures of which are incorporated herein by references.)

A second and less explored strategy for reducing capital intensity is to convert one wafer of semiconductor into a large number of solar cells, thus splitting the capital cost of the wafer over a large number of devices. This strategy may not be viable for silicon because by being an indirect bandgap semiconductor it can require a whole wafer of material to efficiently absorb sunlight. But, for the direct bandgap III-V semiconductors such as GaAs and InP that do strongly absorb sunlight, this may be a viable strategy. Previous work has been done on making GaAs and InP solar cells with 'thin' active layers by using techniques such as epitaxial-lift-off, controlled mechanical spalling, and nanowire-on-wafer. (See, e.g., G. J. Bauhuis, et al., *Sol. Energy Mater. Sol. Cells*, 2009, 93, 9, 1488-1491; E. Yablonovitch, et al., *Appl. Phys. Lett.*, 1990, 56, 24, 2419-2421; the disclosures of which are incorporated herein by references.) However, the active layers of these films are always grown epitaxially, whose enormous capital intensity preclude their mass-manufacture. (See, e.g., A. Goodrich, et al., *A Manufacturing Cost Analysis Relevant to Single and Dual Junction Photovoltaic Cells Fabricated with III-Vs and III-Vs Grown on Czochralski Silicon*, 2013; the disclosure of which is incorporated herein by reference.)

Previous work has focused on non-epitaxial GaAs heterostructures, using PEDOT:PSS and graphene for charge collection, however the largest V_{oc} reported for an ungated device is 780 mV. (See, e.g., L. Yan, et al., *Org. Electron.*, 2015, 16, 71-76; K. Y. Cheng, et al., *IEEE 40th Photovoltaic Specialist Conference, PVSC*, 2014, 1519-1521; X. Li, et al., *Nano Energy*, 2015, 16, 310-319; the disclosures of which are incorporated herein by references.) The wafer-free photovoltaic research has identified easy-to-use charge transport layers and proposed using these layers for charge transport on III-V thin films produced using low-cost techniques such as spalling and nanostructure exfoliation.

Many embodiments provide heterojunction solar cells with high V_{oc} that are fabricated with non-epitaxial methods. Several embodiments implement III-V semiconductors including (but not limited to): n-type doped GaAs and p-type doped GaAs in the heterojunction solar cells. Some embodiments provide a simple, robust, and residue-free GaAs passivation process that results in a heterostructure-grade surface. A number of embodiments provide process conditions for depositing a heterojunction that does not degrade the GaAs surface. Certain embodiments provide GaAs/organic hybrid heterojunction solar cells with V_{oc} of at least 830 mV without epitaxial processing or external doping on a bulk wafer. In some embodiments, the heterojunction solar cells have an area of about 10 mm^2 .

Non-Epitaxial Heterojunction Solar Cell

Many embodiments implement heterojunction including III-V semiconductors solar cells fabricated with non-epitaxial processes. Several embodiments implement passivation processes to create heterostructure-grade surface on the semiconductor substrates. Some embodiments provide electrical contacts to the semiconductor substrates by depositing charge carrier transport layers. FIG. 1 illustrate the structure of a heterojunction solar cell in accordance with an embodi-

ment of the invention. Many embodiments implement heterojunction solar cells (110) with III-V semiconductor materials. Semiconductor materials provide the substrates (102) of heterojunction solar cells. Examples of semiconductor materials include (but are not limited to): n-type doped GaAs (n-GaAs) and p-type doped GaAs (p-GaAs). Some embodiments implement copper, silver, and/or gold as electrodes of the solar cells. The copper and/or silver electrodes can be prepared by thermal evaporation to achieve about 20 nm in thickness. As can readily be appreciated, any of a variety of semiconductor substrates can be utilized as appropriate to the requirements of specific applications in accordance with various embodiments.

In many embodiments, the semiconductor substrates can have back contacts (101) deposited on the back side of the substrates. The back contacts form Ohmic contact with the semiconductor. Examples of back contacts include (but are not limited to): copper, silver, gold, and germanium. As can readily be appreciated, any of a variety of electrodes can be utilized as appropriate to the requirements of specific applications in accordance with various embodiments.

Many embodiments provide non-epitaxial processes to form high Schottky barrier solar cells. Some embodiments passivate heterojunction solar cell surfaces and also make electrical contacts to them to achieve high Schottky barriers. Several embodiments provide that the passivation layers (103) can include organic layers that can bond to the solar cell surfaces and passivate them. In some embodiments, the passivation layers can be organic layers containing at least one thiol group. Examples of thiol containing organic layers include (but are not limited to): octanethiol (OT), 3,3,4,4,5,5,6,6,7,7,8,8,8-tridecafluoro-1-octanethiol (FOT), pentafluorothiophenol (PFBT). The thiol group containing organic layers in accordance with some embodiments can be dissolved in about 5 mL of ethanol and ammonia water mixture (4 parts ethanol and 1-part ammonia water) for about an hour and then rinsed in methanol or isopropyl alcohol (IPA) to passivate the semiconductor substrates.

Several embodiments provide that the passivation layers (103) can be organic layers containing N-heterocyclic-carbene. Examples of N-heterocyclic-carbene containing organic layers include (but are not limited to): 1,3-Diisopropylimidazolium Hydrogencarbonate (NHC). In some embodiments, the N-heterocyclic-carbene containing passivation layers can be formed by mixing NHC with about 50 mg of 1,3-Diisopropylimidazolium Hydrogencarbonate in about 5 mL of 7N ammonia in methanol for about 5 minutes and then rinsed in methanol or IPA.

The passivation layers (103) in accordance with certain embodiments can be organic layers containing sulfide ions including (but not limited to) ammonium sulfide. Several embodiments provide that equal volume mixture of ammonium sulfide, ammonia water, and hydrazine monohydrate can be mixed with about 5 wt % elemental sulfur at 50° C. for one minute then rinsed under DI water to form the passivation layers.

In a number of embodiments, the passivation layers (103) can include dithiothreitol (DTT), and cysteine (Cys). In many embodiments, about 50 mg of cysteine or dithiothreitol can be dissolved in about 5 mL of ethanol and ammonia water mixture (4 parts ethanol and 1-part ammonia water) for about an hour and then rinsed in methanol or IPA to form the passivation layers. As can readily be appreciated, any of a variety of passivation layers can be utilized as appropriate to the requirements of specific applications in accordance with various embodiments.

In some embodiments, electron and/or hole transport contacts (104) can be deposited on top of the passivation layers to protect the passivation layers and the solar cell surfaces. In several embodiments, the electron/hole contact layers contain bulky tails with a certain length so that electrons and/or holes can pass through. Certain embodiments provide that electron and/or hole transport contact layers should not destroy the semiconductor and the passivation layer interface. In a number of embodiments, electron and/or hole transport contacts can be solution deposited nanoparticles, solution deposited organics, evaporated metal oxides, and organics. Some embodiments provide that the organics may allow deposition of metals to make high-barrier Schottky contacts.

Several embodiments provide that hole transport layers (104) can include (but are not limited to): poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS), 2,2',7,7'-Tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (SPIRO), SPIRO doped with 5 wt % Tris(pentafluorophenyl)borane, tris(4-carbazoyl-9-ylphenyl)amine (TCTA), TCTA doped with 5 wt % Tris(pentafluorophenyl)borane, polystyrene (PS), 4,4'-Bis(N-carbazoyl)-1,1'-biphenyl (CBP), copper (I) iodide (CuI) dissolved in acetonitrile, polyvinyl alcohol (PVA) dissolved in water, copper thiocyanate dissolved in ammonia water (CuSCN), 1,1-Bis[(di-4-tolylamino)phenyl]cyclohexane (TAPC), TAPC doped with 5 wt % Tris(pentafluorophenyl)borane, poly(triaryl amine) (PTAA), PTAA doped with 5 wt % Tris(pentafluorophenyl)borane, nickel oxide (NiO) nanoparticles, NiO by electron beam evaporation deposition. In some embodiments, the PEDOT:PSS hole transport layer can be mixed with about 10 wt % DMSO and spun coat at about 3000 rpm for about 30 seconds, then baked on a hot plate at about 130° C. for about 5 minutes. Certain embodiments provide that organic hole transport layers can be dissolved in chlorobenzene and then spun coat to achieve about 10 nm in thickness. Electron beam evaporation deposited NiO layers in accordance with certain embodiments can be prepared by electron beam evaporation of nickel with oxygen at a pressure of about 5E-5 Torr. As can readily be appreciated, any of a variety of hole transport layers can be utilized as appropriate to the requirements of specific applications in accordance with various embodiments.

Electron transport layers (104) in accordance with many embodiments can include (but are not limited to): tin oxide (SnO₂) nanoparticles, aluminum doped zinc oxide (AZO) nanoparticles, C60 (buckyballs). In several embodiments, nanoparticles layers can be prepared from nanoparticle dispersions from Avantama®. In some embodiments, C60 layers can be prepared by thermal evaporation to achieve about 10 nm in thickness. As can readily be appreciated, any of a variety of electron transport layers can be utilized as appropriate to the requirements of specific applications in accordance with various embodiments.

In a number of embodiments, electrode modified layers (105) can be deposited on top of electron/hole transport layers. The electrode modified layers in accordance with several embodiments can set the work function of the contacts. Examples of the electrode modified layers for n-GaAs include (but are not limited to): tungsten trioxide (WO₃), molybdenum trioxide (MoO₃). Examples of electrode modified layers for p-GaAs include (but are not limited to): lithium fluoride (LiF), caesium oxide (Cs₂O). As can readily be appreciated, any of a variety of electrode modified layers can be utilized as appropriate to the requirements of specific applications in accordance with various embodiments.

In several embodiments, lateral current transport layers (106) can be deposited on top of the contacts. The lateral current transport layers in accordance with some embodiments can be transparent conductors. Examples of transparent conducting layers include (but are not limited to): poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS), indium tin oxide (ITO). Certain embodiments include silver nanowires in the lateral transport layers. The lateral current transport layers can be optional. As can readily be appreciated, any of a variety of lateral current transport layers can be utilized as appropriate to the requirements of specific applications in accordance with various embodiments.

Many embodiments provide that open-circuit voltage (V_{oc}) of high Schottky barrier heterojunction solar cells can range from about 20 mV to about 900 mV. In some embodiments, V_{oc} can be about 830 mV. The efficiency of the heterojunction solar cells in accordance with several embodiments can be at least 20%. Table 1 lists V_{oc} of n-GaAs organic heterojunction solar cells with various combinations of passivation layers, hole transport layers (HTL), electron transport layers (ETL), and electrode modifier layers.

TABLE 1

V_{oc} of n-GaAs heterojunction solar cells with different passivation layers, HTL, ETL, and electrode modifier layers.				
Passivant	HTL	ETL	Electrode Modifier	V_{oc} , mV
OT		AZO NP		20
NHC	CBP		MoO ₃	115
DTT	CuI		MoO ₃	120
DTT	TCTA Doped		MoO ₃	154
DTT	TAPC		MoO ₃	172
DTT	TAPC Doped		MoO ₃	176
NHC	PS		MoO ₃	205
DTT	CBP		MoO ₃	244
NHC	TCTA Doped		MoO ₃	269
OT		SnO ₂ NP		281
DTT	CuSCN		MoO ₃	305
NHC	CuSCN		MoO ₃	312
OT	NiO EB			316
DTT	PTAA Doped		MoO ₃	327
NHC	TAPC		MoO ₃	337
NHC	TAPC		MoO ₃	337
NHC	TCTA		MoO ₃	340
NHC	TAPC Doped		MoO ₃	340
NHC	NONE		MoO ₃	342
Sulfide	NiO EB			363
DTT			MoO ₃	386
DTT	PS		MoO ₃	389
DTT	NiO NP		MoO ₃	390
NHC	PTAA Doped		MoO ₃	395
OT			MoO ₃	404
Cys			MoO ₃	405
DTT	PS		MoO ₃	451
FOT	NiO NP		MoO ₃	458
OT		C60		468
DTT	TCTA		MoO ₃	492
Sulfide	TCTA		WO ₃	511
DTT	PVA		MoO ₃	521
FOT	TCTA		MoO ₃	523
FOT	NONE		MoO ₃	545
NHC	CBP		MoO ₃	560
Sulfide	PVA		MoO ₃	588
DTT	PEDOT		MoO ₃	594
Sulfide	TAPC		MoO ₃	602
NHC	PS		MoO ₃	609
OT	NiO NP			610
DTT	TCTA		MoO ₃	611
Sulfide	TCTA Doped		MoO ₃	620
FOT	PVA		MoO ₃	621
Sulfide	PS		MoO ₃	622
Sulfide	TAPC Doped		MoO ₃	632

TABLE 1-continued

V_{oc} of n-GaAs heterojunction solar cells with different passivation layers, HTL, ETL, and electrode modifier layers.				
Passivant	HTL	ETL	Electrode Modifier	V_{oc} , mV
PFBT	NONE		MoO ₃	634
PFBT	CuSCN		MoO ₃	639
Sulfide	TCTA		MoO ₃	664
DTT	PTAA		MoO ₃	670
Sulfide	SPIRO Doped		MoO ₃	670
PFBT	TCTA		MoO ₃	682
Sulfide	SPIRO		MoO ₃	685
Sulfide	PEDOT			719
DTT	PS		MoO ₃	721
Sulfide	PTAA Doped		MoO ₃	725
Sulfide	PTAA		MoO ₃	731
Sulfide	NiO NP			747
PTAA			MoO ₃	752
Sulfide	SPIRO			773
Sulfide	NiO NP		WO ₃	793
Sulfide	TCTA		WO ₃	842

GaAs/TCTA/WO₃ Heterojunction Solar Cells

Many embodiments provide GaAs heterojunction solar cells that can achieve high V_{oc} without any epitaxial processing or external doping on a bulk wafer. In some embodiments, the GaAs heterojunction solar cells have an area of about 10 mm². Several embodiments provide the effect of various thickness of hole selective contact layers including (but not limited to) TCTA on device performance. A structure of GaAs/TCTA/WO₃ heterojunction solar cell in accordance with an embodiment of the invention is illustrated in FIG. 2. The semiconductor GaAs substrate (211) can be a n-type GaAs with about 5E17 doping density. The GaAs substrate can be about 500 μ m in thickness. On the backside of the GaAs substrate, a back contact (210) can be deposited. The back contact can be a CuGe Ohmic contact. On the top side of the GaAs substrate, a passivation layer (212) can be deposited to passivate the semiconductor surface. The passivation can be a sulfide passivated layer of GaS monolayer. A hole selective contact layer (213) that transports hole charge carriers can be coated on the passivation layer. The hole selective layers can be TATA with variable thickness from about 5 nm to about 30 nm. An electrode modified layer (not shown) can be deposited on the hole contact layer to set the work function. The electrode modified layer can be a WO₃ film of about 5 nm in thickness. A lateral transport layer (214) including (but not limited to): PEDOT:PSS, can be deposited to improve charge transport.

The solar cell with a thinner TCTA hole contact achieves a higher V_{oc} in accordance with many embodiments. IV curves of GaAs/TCTA/WO₃/PEDOT:PSS heterojunction solar cell with TCTA at 5 nm and 30 nm thickness in accordance with an embodiment are illustrated in FIGS. 3A and 3B respectively. FIG. 3A shows the IV curve of the solar cell with a Cu/Ge Ohmic back contact, a 500 μ m n-GaAs substrate, a sulfide passivated GaS monolayer, a 5 nm TCTA, a 5 nm WO₃, and a PEDOT:PSS lateral transport layer. The device has V_{oc} of about 840 mV. FIG. 3B shows the IV curve of the solar cell with a Cu/Ge Ohmic back contact, a 500 μ m n-GaAs substrate, a sulfide passivated GaS monolayer, a 30 nm TCTA, a 5 nm WO₃, and a PEDOT:PSS lateral transport layer. The device has V_{oc} of about 806 mV.

Several embodiments provide an optimal TCTA thickness that can achieve a highest V_{oc} in GaAs/TCTA/WO₃/PEDOT:PSS heterojunction solar cells. A graph of V_{oc} at various TCTA thickness in accordance with an embodiment of the

invention is illustrated in FIG. 4. FIG. 4 shows that at a TCTA thickness of about 5-6 nm, V_{oc} of GaAs/TCTA/ WO_3 /PEDOT:PSS heterojunction solar cells achieves a value of about 840 mV.

Many embodiments provide performances of GaAs/TCTA/ WO_3 /PEDOT:PSS heterojunction solar cells with an optimal TCTA thickness. Some embodiments include a bulk, n-type GaAs wafer with sulfur passivation, 5 nm or 10 nm of spun coat TCTA, an organic p-type semiconductor, 5 nm of evaporated WO_3 , and a spun-coat film of highly-conductive PEDOT:PSS for lateral current transport. Certain embodiments provide that the GaAs/organic heterojunction solar cells can achieve about 830 mV V_{oc} . Heterojunction solar cells with about 830 mV V_{oc} in accordance with certain embodiments can achieve about 20% efficiency with low-capital intensity processing.

Many embodiments describe passivation processes to produce the GaAs organic heterojunction solar cells. The passivation processes are described in details in the Exemplary Embodiments. The passivation can be characterized by time-resolved photoluminescence (TRPL), where TRPL lifetimes can be calculated as a function of barrier height and then compared to the measured lifetime. The passivation layers can also be characterized by reflection high energy electron diffraction (RHEED). The passivation layers can achieve lifetimes of about 1.2 ns. The TRPL of an unpassivated sample is below the noise-floor of the instrument. The quality of the GaAs surface can be monitored with the TRPL after each step during the device formation process. After TCTA and WO_3 deposition the lifetime improves to about 1.4 ns, and drop to about 0.7 ns after PEDOT:PSS application. The final lifetime can be used to estimate the achievable open-circuit voltage by using it and the device thickness to calculate the steady-state carrier concentration and thus the quasi-Fermi level splitting using the relation:

$$V_{oc} = \frac{kT}{q} \ln \left(\frac{N_D \tau J_L}{q d n_i^2} \right), \quad (1)$$

where kT is the thermal voltage (26 meV), q is the elementary charge, N_D is the donor concentration, J_L is the photocurrent density, assumed to be 30 mAcm^{-2} , d is the device thickness, and n_i the intrinsic carrier concentration. The TRPL implies that V_{oc} is about 950 mV. To further characterize the GaAs surface, RHEED can be performed after passivation to determine the nature of the surface structure. Before loading into the RHEED system, the sample is annealed at about 300° C. in air for about 5 minutes to sublimate any sulfur on the surface. The RHEED pattern is illustrated in FIG. 5 in accordance with an embodiment. The distinct diffraction spots indicate the surface is structured as opposed to disordered. From the distinct diffraction lines, some embodiments provide that the GaAs surface is structured, even after having been annealed in air. The lifetime of the annealed sample is about 0.9 ns. The maintaining of an ordered structure and about 1 ns lifetime after a variety of processing in ambient illustrate the robustness of the passivation process in accordance with certain embodiments.

Some embodiments provide the electrical characterizations for GaAs organic heterojunction solar cells. The IV curve of the GaAs/TCTA/ WO_3 /PEDOT:PSS heterojunction solar cell in accordance with an embodiment of the invention is illustrated in FIG. 6. The solar cell achieves V_{oc} of about 830 mV. The cell is currently limited by large series resistance and low J_{sc} of about 10.6 mAcm^{-2} . The series

resistance can be due to the sub-optimal contact design of using small dots many millimeters from each other. A saturating current in reverse bias can be observed. The low J_{sc} may be due to the wafer doping which limits the depletion width, and thus charge collection depth, to less than 100 nm. A lower-doped wafer should remedy the low J_{sc} .

The dark IV curve of the GaAs/TCTA/ WO_3 /PEDOT:PSS heterojunction solar cell in accordance with an embodiment of the invention is illustrated in FIG. 7. The dark IV curve is a diode with ideality factor of about 1.85 and J_0 of about $10^{-7} \text{ mAcm}^{-2}$. Using superposition and assuming short circuit current to be about 29 mAcm^{-2} , this device would be about 20.5% efficient. Many embodiments provide processes that allow the formation of high-quality, air-exposed GaAs surfaces that enable solar cells produced with simple processing to achieve V_{oc} that can compete with commercial silicon devices. Some embodiments provide that such devices can produce about 830 mV V_{oc} .

GaAs/Thiol/ WO_3 Heterojunction Solar Cells

Many embodiments provide high-barrier Schottky GaAs heterojunction solar cells that can achieve high V_{oc} without any epitaxial processing or external doping on a bulk wafer. A structure of GaAs/Thiol/ WO_3 heterojunction solar cell in accordance with an embodiment of the invention is illustrated in FIG. 8. The semiconductor GaAs substrate (**811**) can be a n-type GaAs with about $5E16$ doping density. The GaAs substrate can be about 10 μm in thickness. On the backside of the GaAs substrate, a back contact (**810**) can be deposited. The back contact can be a CuGe Ohmic contact. On the top side of the GaAs substrate, a passivation layer (**812**) can be deposited to passivate the semiconductor surface. The passivation can be a thiol containing layer including (but not limited to): OT, Cys, and DTT. A hole selective contact layer (**813**) that transports hole charge carriers can be coated on the passivation layer. The hole selective layers can be WO_3 film of about 30 nm in thickness. The thiol containing passivation layer can protect the GaAs from high work function WO_3 to form a high-barrier Schottky junction. A lateral transport layer (**814**) including (but not limited to): silver nanowires, can be deposited to improve charge transport. The Schottky barrier height of the GaAs/Thiol/ WO_3 heterojunction solar cell measured by TRPL is about 1.38 eV. The Schottky barrier height implies a solar cell with a V_{oc} of about 905 mV, and the device would be about 21.5% efficient. Table 2 lists Schottky barrier height and V_{oc} of GaAs/Thiol/ WO_3 heterojunction solar cell with various thiol containing passivation layer.

TABLE 2

Schottky barrier height and V_{oc} of GaAs/Thiol/ WO_3 heterojunction solar cell			
Passivant	OT	Cys	DTT
Thiol τ , ns	2.1	1.9	2.1
Thiol/ WO_3 τ , ns	0.58	0.67	0.54
Barrier, eV	1.34	1.38	1.33
Implied V_{oc} , eV	0.90	0.91	0.90

EXEMPLARY EMBODIMENTS

Although specific embodiments of compositions, methods and process are discussed in the following sections it will be understood that these embodiments are provided as exemplary and are not intended to be limiting.

Many embodiments implement two types of GaAs wafers. First one is 100, n-GaAs wafers with $5 \times 10^{17} \text{ cm}^{-3}$ carrier concentration. These are used for making solar cells. Second one is n-GaAs wafers with a 50 nm GaInP layer and 1 μm GaAs layer grown on top, both with $3 \times 10^{16} \text{ cm}^{-3}$ carrier concentration. The wafers with epitaxial layers are used for optimizing the passivation and heterojunction formation procedures because the GaInP uniformly confines the photogenerated holes to the top GaAs layer which causes the time resolved photoluminescence (TRPL) signal to be an easy to interpret exponential decay whose lifetime indicates the surface quality.

TRPL is measured using a 405 nm Picoquant pulsed laser running just above threshold for photoexcitation, and then a single photon detector and time correlator are used for extracting the TRPL signal.

For electrically active devices, back contacts to the GaAs are made by sequentially sonicating the GaAs chip in acetone and isopropanol for three minutes each, followed by a 30 second dip in concentrated HCl to etch the native oxide, and then evaporating 100 nm of Cu followed by 50 nm of Ge without breaking vacuum, and finally annealing at 400°C . in a nitrogen purged tube furnace for 5 minutes. A CuGe contact over the AuGeNi because of the lower formation temperature and the lack of spiking.

For passivation, chips are first sonicated as above, and then an equivolume mixture of saturated ammonium hydroxide, ammonium sulfide, hydrazine monohydrate is used, with a further added 5% elemental sulfur by mass. The hydroxide, sulfide, and sulfur form a corrosive mixture that slowly etches the GaAs and terminates it with sulfur. The hydrazine may be needed for preventing residue formation. The passivation is performed by putting a small dish of water and the chips on a hotplate at 40°C ., dropping the sulfide solution on the chips, and then covering the dish and chips with a petri dish for 30 minutes, followed by immediate rinsing with flowing DI water and then drying under flowing nitrogen. The small dish of water can prevent the passivation solution on the chips from drying which would result in a difficult to remove deposit on the chips. Passivation can be performed after back contact formation for electrically active devices. The process may corrode the back contacts, but they are still satisfactorily ohmic.

Thiol passivation can be formed by treatment with octanethiol, cysteine, or dithiothreitol in ethanol and ammonia solution at about 40°C . overnight.

To form the heterojunction, the wafer is first coated with TCTA by spin-coating a 0.3% by weight solution in chlorobenzene at 3000 rpm for 30 s followed by annealing on a hot plate at 130°C . for 1 minute, resulting in a 10 nm thick film. Then a thermally evaporated 5 nm of WO_3 is deposited at 0.2 A/s to establish a deep work function at the surface of the TCTA. Next, PEDOT:PSS doped with 10% isopropanol, 10% DMSO, 10% imidazole, and 0.1% CTAB, all by mass, is spun coat at 2000 rpm for 30 s, followed by drying on a hotplate at 130°C . for 5 minutes. The isopropanol and CTAB aid with wetting and coating, the DMSO enhances the conductivity, and the imidazole reduces the PEDOT's corrosivity. The films have about with roughly $500 \Omega/\square$ sheet resistance, as measured by a four-point probe.

Top metal contacts are 100 nm thick gold squares defined by 0.01 mm^2 square holes in a shadow-mask deposited by electron-beam evaporation at 2 A/s. Edge isolation is done with a razor blade. Cells are characterized using an AM 1.5 g solar simulator calibrated with a silicon reference cell.

This description of the invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form described, and many modifications and variations are possible in light of the teaching above. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications. This description will enable others skilled in the art to best utilize and practice the invention in various embodiments and with various modifications as are suited to a particular use. The scope of the invention is defined by the following claims.

The invention claimed is:

1. A heterojunction solar cell, comprising:

at least one back contact;

at least one semiconductor substrate, wherein the at least one semiconductor substrate is an n-type doped GaAs or a p-type doped GaAs;

at least one passivation layer, wherein the at least one passivation layer passivates one surface of the at least one semiconductor substrate and comprises one of: at least one thiol group, 1,3-Diisopropylimidazolium Hydrogencarbonate (NHC), or at least one sulfide ion;

at least one contact layer, wherein the at least one contact layer is a hole transport layer or an electron transport layer, and the contact layer is on top of the at least one passivation layer; wherein the hole transport layer comprises one of poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS), 2,2',7,7'-Tetrakis [N,N-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (SPIRO), SPIRO doped with 5 wt % Tris(pentafluorophenyl)borane, tris(4-carbazoyl-9-ylphenyl) amine (TCTA), TCTA doped with 5 wt % Tris (pentafluorophenyl)borane, polystyrene (PS), 4,4'-Bis (N-carbazoyl)-1,1'-biphenyl(CBP), copper (I) iodide (CuI) dissolved in acetonitrile, polyvinyl alcohol (PVA), copper thiocyanate (CuSCN), 1,1-Bis[(di-4-tolylamino)phenyl]cyclohexane (TAPC), TAPC doped with 5 wt % Tris(pentafluorophenyl)borane, poly(triaryl amine) (PTAA), PTAA doped with 5 wt % Tris (pentafluorophenyl)borane, nickel oxide (NiO) nanoparticles, or NiO film; and

at least one electrode modified layer, wherein the at least one electrode modified layer is on top of the at least one contact layer;

wherein the at least one passivation layer is positioned between the at least one semiconductor substrate and the at least one contact layer and bonds to both the at least one semiconductor substrate and the at least one contact layer;

wherein the open circuit voltage of the solar cell is at least 830 mV; and

wherein the heterojunction solar cell is non-epitaxial.

2. The non-epitaxial heterojunction solar cell of claim 1, wherein the at least one back contact is an ohmic contact comprising Cu and Ge.

3. The non-epitaxial heterojunction solar cell of claim 1, wherein the organic film comprises octanethiol (OT), 3,3,4,4,5,5,6,6,7,7,8,8,8-tridecafluoro-1-octanethiol (FOT), pentafluorothiophenol (PFBT), dithiothreitol (DTT), or cysteine (Cys).

4. The non-epitaxial heterojunction solar cell of claim 1, wherein the at least one passivation layer comprises ammonium sulfide.

5. The non-epitaxial heterojunction solar cell of claim 1, wherein the electron transport layer comprises one of tin

oxide (SnO_2) nanoparticles, aluminum doped zinc oxide (AZO) nanoparticles, or C60.

6. The non-epitaxial heterojunction solar cell of claim 1, wherein the at least one semiconductor substrate is an n-type doped GaAs and the at least one electrode modified layer 5 comprises tungsten trioxide (WO_3) or molybdenum trioxide (MoO_3).

7. The non-epitaxial heterojunction solar cell of claim 1, wherein the at least one semiconductor substrate is a p-type doped GaAs and the at least one electrode modified layer 10 comprises lithium fluoride (LiF) or caesium oxide (Cs_2O).

8. The non-epitaxial heterojunction solar cell of claim 1, further comprising a lateral current transport layer, wherein the lateral current transport layer comprises poly (3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT: 15 PSS), indium tin oxide (ITO), or silver nanowires.

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